

Clustering of Song Lyrics: A Comparison of Clustering Methods with Sentence-BERT

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Abstract—This study explores the semantic clustering of song lyrics using Sentence-BERT and three families of clustering algorithms: Agglomerative (hierarchical), K-Means (partitioning), and HDBSCAN (density-based). Guided by the CRISP-DM methodology, we collected and preprocessed a dataset of 10,000 lyrics scraped from genius.com, filtered for linguistic consistency and thematic relevance. By applying SBERT for vectorization, we aim to uncover latent groupings in lyrical content and assess the effectiveness of clustering algorithms in producing semantically coherent clusters. Our results show that HDBSCAN delivers the strongest internal cohesion, achieving a silhouette score of approximately 0.63 with two automatically inferred clusters. While Agglomerative and K-Means both perform optimally at a binary partition ($k = 2$), yielding more modest silhouette scores (~ 0.04 and ~ 0.06 , respectively), extending to three clusters uncovers interpretable subthemes but leads to diminishing overall validity.

Keywords: Song Lyrics, Semantic Clustering, CRISP-DM, Sentence-BERT, Agglomerative, K-Means, HDBSCAN.

I. INTRODUCTION

Clustering song lyrics is vital in music information retrieval, supporting applications like personalized recommendations, cultural analysis, and thematic cataloging. As digital music libraries grow, manual tagging becomes less scalable and more prone to cultural bias [1]. Clustering provides an objective alternative, grouping songs by semantic themes or stylistic traits, benefiting both users and curators [2].

This study follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework, which structures data mining into six iterative phases: business understanding, data understanding, data preparation, modeling, evaluation, and deployment. Each phase offers clear guidance and outputs, promoting reproducibility and informed decision-making. In this context, CRISP-DM ensures methodological rigor across preprocessing, clustering, and evaluation stages, allowing a transparent comparison of hierarchical, partitioned, and density-based algorithms [3][4].

The research is guided by the following questions:

RQ1. Comparative Algorithm Performance: How do Agglomerative, K-Means, and HDBSCAN differ in generating semantically coherent lyric clusters?

RQ2. Impact of Hyper-parameter Configuration: Which hyperparameter settings yield optimal intrinsic clustering scores?

RQ3. Emergence of Genre or Thematic Groupings: Can unsupervised clustering uncover latent themes or genres that align with recognizable musical categories?

These questions align with CRISP-DM phases: RQ3 supports the business and analytical understanding, RQ1 addresses data exploration and modeling, RQ2 focuses on model tuning, and all RQs guide the evaluation phase.

II. RELATED WORK

Previous studies on clustering song lyrics have explored various methods, highlighting different approaches and their respective limitations. Masdha Radio (2021) utilized K-means with variance initialization, a method that, while effective in some cases, is constrained by its reliance on a single algorithm and is limited to Indonesian lyrics [5]. Meanwhile, the CokeStudio Analysis (2024) took a multi-algorithm comparison approach, offering a broader scope but focusing primarily on genre-specific (Pakistani music) data, which limits its applicability to other genres [2]. These studies illustrate a variety of strategies for lyric clustering but underscore the challenges of algorithmic limitations, data specificity, and the cost of computational resources which struggle to capture sentence-level semantics and contextual relationships.

SBERT addresses these gaps by producing context-aware sentence embeddings optimized for semantic similarity. Unlike BERT's token-level outputs, SBERT's siamese architecture fine-tunes sentence pairs with triplet loss, directly aligning embeddings with human-like semantic judgment. This enables detection of latent themes (e.g., distinguishing romantic vs. destructive themes in "fire"-related lyrics) that word-based models oversimplify. Furthermore, SBERT's superiority was demonstrated in semantic tasks, achieving Spearman correlations of > 0.80 on STS benchmarks far exceeding word-embedding baselines. This validates its use for identifying culturally or thematically coherent song clusters. This revision emphasizes three key SBERT advantages: contextual precision, computational efficiency, and task-specific optimization [6]. This research builds upon these works by adopting a more generalizable approach using hierarchical,

partitioned, and density-based algorithms, aiming to overcome the constraints highlighted in previous studies.

III. BUSINESS AND DATA UNDERSTANDING

A. Problem Background and Business Motivation

Along with the rise of digital music consumption, user preferences in genre and artist are shifting into more contextual aspects, such as mood, theme, or lyrical meaning. Most of today's music streaming platforms have a music recommendation system. The recommendation system remains limited to filtering based on metadata, without analyzing deep into the lyrics content. This presents an opportunity to explore content-based approaches that utilize text mining to extract meaningful patterns from lyrics, enabling a more personalized recommendation system.

B. Business Objective and Stakeholders

The primary objective of this project is to cluster songs based on lyrical similarity using various text clustering techniques. This approach aims to uncover thematic or emotional patterns contained in songs. In addition, this project aims to analyze the correlation between lyric content, release year, and song popularity (in terms of number of views), offering insight into lyrical trends over time. Key stakeholders include music streaming users, recommendation system developers, and music industry professionals.

C. Expected Impact and Risks

The implementation of lyrics-based recommendation has the potential to provide a richer and more personal experience in song searching. However, there are some risks and challenges to consider: limitations in data scraping such as broken URLs or inaccessible pages on genius.com, unstructured lyrics format, and uneven data distribution

D. Dataset Overview

The dataset used in this study was collected via web scraping from genius.com and comprises a total of 10,000 songs, split into a training set of 9,000 songs and a testing set of 1,000 songs. Each entry contains the following attributes: title (the song title), artist (the name of the performer or group, year (year of release), views (number of listeners or view count, serving as a proxy for popularity, URL (web address linking to the lyrics page), lyric (the textual content of the song's lyrics).

E. Data Quality and Initial Exploration

Preliminary analysis confirmed no duplicate entries, ensuring each song record is unique. However, the training set had 1,633 missing values and the test set 189, mostly due to failed scrapes or unavailable lyrics. Exploratory analysis included a word cloud to highlight frequently used terms, offering early insight into dominant themes and emotional tones. A temporal analysis showed publication trends across years, revealing fluctuations in music production. Additionally, certain artists stood out by song frequency and total views, indicating their strong influence on popular lyrical content. These findings support the potential for meaningful pattern discovery through clustering.

F. Data Preparation

The dataset used in this study consists of several

columns: title, artist, year, view, url, and lyric. For the analysis, the lyric column was the primary focus for the clustering process, while the other columns were utilized for data exploration and interpretation.

Initial analysis revealed several missing lyric entries due to unsuccessful scraping attempts from genius.com. Entries with completely unavailable lyrics were removed, resulting in 7,367 training and 811 testing entries. Further filtering was conducted to retain only linguistically consistent data. Using the langdetect library, non-English lyrics were excluded. Additionally, lyrics that were too short or lacked meaningful content were discarded. The final datasets consisted of 7,303 training and 808 testing entries. To address cases where lyrics were not found due to dynamic HTML structure, an alternative scraping method was used to recover text from previously unresolved URLs. This significantly reduced data loss and improved coverage.

Text preprocessing included expanding contractions (e.g., "can't" → "cannot"), converting text to lowercase, and removing punctuation and numeric characters. Tokenization was performed using NLTK's word_tokenize, and words with exaggerated letter repetitions were normalized (e.g., "sooo" → "soo"). Lemmatization using WordNet was applied to standardize words to their base form, and common stop words were removed to reduce noise. Tokens shorter than three characters were excluded to eliminate low-information words.

Earlier vectorization techniques such as Bag of Words and TF-IDF were explored but ultimately replaced with contextual embeddings using the all-MiniLM-L6-v2 model from SentenceTransformers. This model produced 384-dimensional vectors for each lyric, enabling semantic clustering at the sentence level. The preprocessed datasets were serialized into pickle files to ensure consistent reuse and efficient loading during the modeling phase.

IV. METHODOLOGY

To determine the appropriate number of clusters, the process began with exploratory analysis using a dendrogram generated via Ward's linkage, which guided the initial cluster range selection. For KMeans, the Elbow Method was used alongside internal metrics such as Silhouette Score and Calinski-Harabasz Index to evaluate cluster compactness and separation. Agglomerative Clustering was tested using various affinity-linkage combinations and evaluated using Silhouette, Calinski-Harabasz, and Davies-Bouldin scores. HDBSCAN, which does not require specifying the number of clusters, was tuned across different values of min_cluster_size and min_samples, producing a flexible range of cluster outputs. All models were trained and evaluated on both training and test sets. Metric-based evaluation and qualitative visualization were then used to select the most meaningful clustering configurations for further analysis.

V. MODELING AND EVALUATION

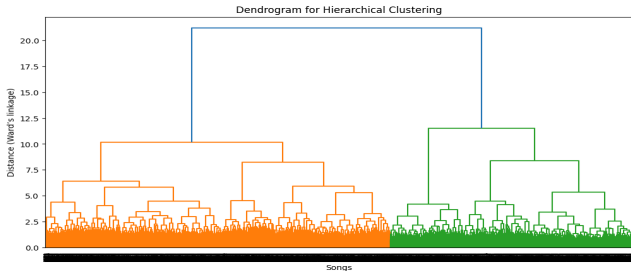


Figure 1

Figure 1 shows the dendrogram from hierarchical clustering using Ward's linkage, with a visible separation at a linkage distance of ~ 8 . This guided the selection of $k = 2$ and $k = 3$ for both K-Means and Agglomerative clustering, further validated using internal metrics such as the Elbow Method, Silhouette Score, Calinski-Harabasz Index, and Davies-Bouldin Index. For HDBSCAN, hyperparameter tuning was performed based on cluster persistence and handling of noise, aiming for a balance between cohesion and granularity.

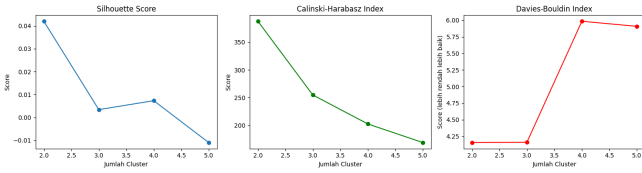


Figure 2

Evaluation metrics for Agglomerative clustering across $k = 2$ to 5 show that $k = 2$ consistently yields the highest Silhouette and Calinski-Harabasz scores, and the lowest Davies-Bouldin index. While $k = 3$ remains acceptable, it introduces higher fragmentation.

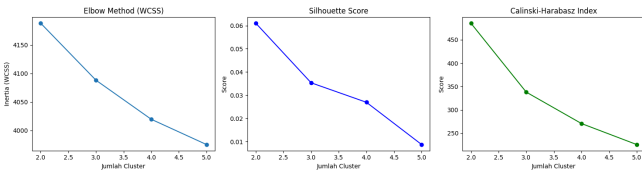


Figure 3

K-Means results support this: the Elbow plot suggests diminishing returns after $k = 3$, while Silhouette and Calinski-Harabasz scores peak at $k = 2$, confirming that a binary partition is the most coherent.

Table 1 summarizes Agglomerative clustering scores across linkage-affinity pairs. The optimal setting for a two-cluster structure is Ward linkage with Euclidean distance, while a three-cluster configuration is best achieved with complete linkage and cosine affinity, allowing finer thematic separation.

n_clusters	linkage	affinity	Silhouette Score	Davies-Bouldin Score	Calinski-Harabasz Score
2	average	cosine	0.378191	0.664109	2.226167
3	average	cosine	0.332920	1.968992	5.334872
3	complete	cosine	0.096213	4.778137	157.643448

2	complete	cosine	0.093527	5.037171	275.829865
2	ward	euclidian	0.042013	4.156001	387.728851
3	ward	euclidian	0.003421	4.159800	254.693695

Table 1

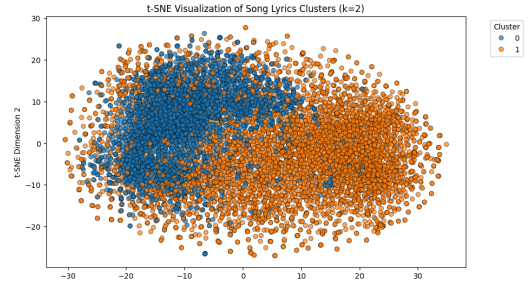


Figure 4

The t-SNE projection of the training set using $k = 2$, Ward-Euclidean reveals a clean left-right cluster division, with each cluster occupying a distinct region and preserving neighborhood structures.

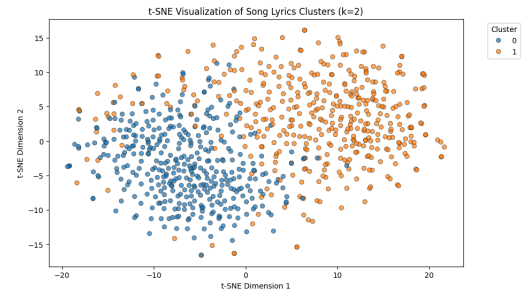


Figure 5

A similar pattern appears in the test set, confirming that the model generalizes well and retains thematic consistency on unseen data.

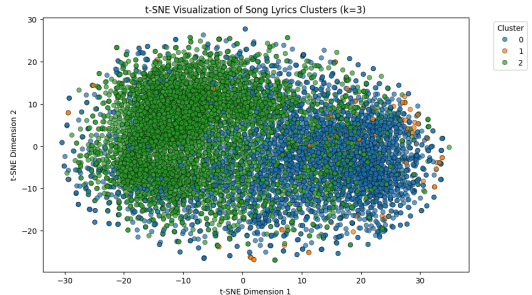


Figure 6

For $k = 3$ with complete-cosine, training embeddings split into three adjacent clusters: one narrative-driven, one motif-heavy, and a transitional group combining both. This indicates that increasing k reveals more nuanced lyrical themes.

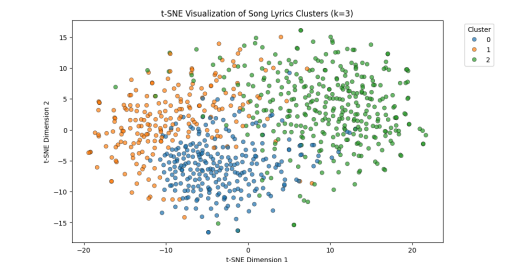


Figure 7

The test set under this setting reproduces the three-cluster layout with reasonable separation, validating the generalizability of the ternary partition.

n_clusters	init_method	WCSS (Inertia)	Silhouette Score	Calinski-Harabasz Score
2	k-means++	4188.0806	0.0610	485.4609
2	random	4188.0806	0.0610	485.4609
3	random	4088.0410	0.0357	337.9604
3	k-means++	4088.0410	0.0354	337.9572

Table 2

The table compares K-Means configurations. The best-performing model is $k = 2$ with k-means++ initialization, which achieves the highest Silhouette (0.0610) and Calinski-Harabasz (485.46) scores. Among $k = 3$ options, random initialization provides slightly lower scores but enables exploration of finer clusters.

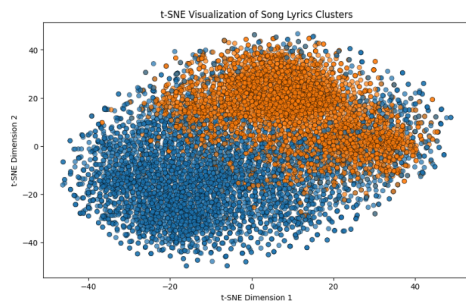


Figure 8

K-Means clustering with $k = 2$ on the training set shows two clearly separated regions, reflecting distinct lyrical themes.

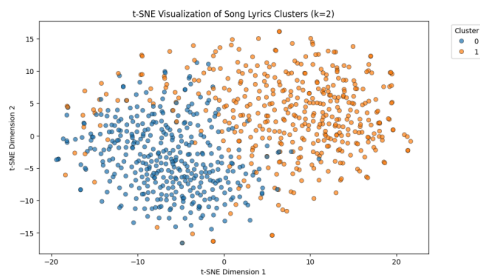


Figure 9

The same model applied to the test set yields similarly distinct clusters, though with slightly more overlap, indicating strong but slightly softer generalization compared to Agglomerative.

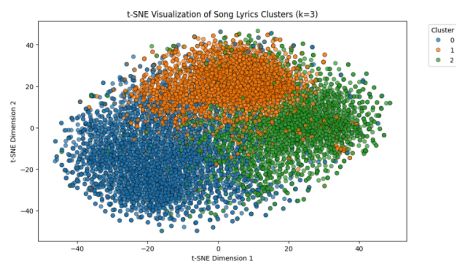


Figure 10

Using $k = 3$, the test set reveals three separate regions with some boundary overlaps, suggesting the model captures distinct yet intersecting themes.

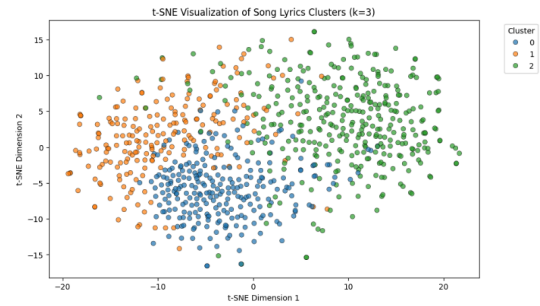


Figure 11

Another $k = 3$ configuration shows a compact lower-left cluster, a center-left group, and a right-dominant cluster—maintaining spatial coherence despite looser separation on the test set.

HDBSCAN was tuned using grid search across min_cluster_size (50–140) and min_samples (50–140), evaluated via Silhouette Score. Best results were obtained with min_cluster_size = 90, min_samples = 120, achieving a Silhouette Score of 0.627 for two clusters. For three clusters, min_cluster_size = 75, min_samples = 140 yielded a Silhouette of 0.493, indicating reasonable but less distinct separation.

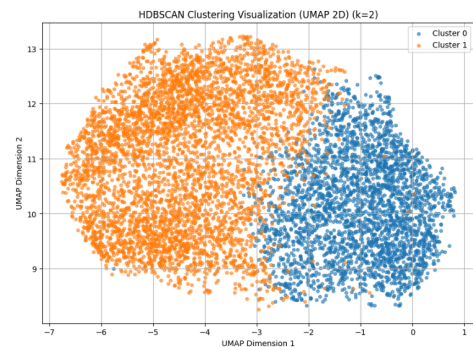


Figure 12

UMAP projection of the training set using the best HDBSCAN configuration shows a sharp left-right separation, with clusters clearly differentiated in the embedding space.

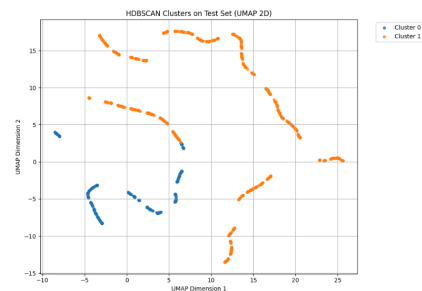


Figure 13

On the test set, cluster structure remains consistent, though more diffuse. This suggests slightly reduced confidence in unseen data but overall stable model behavior.

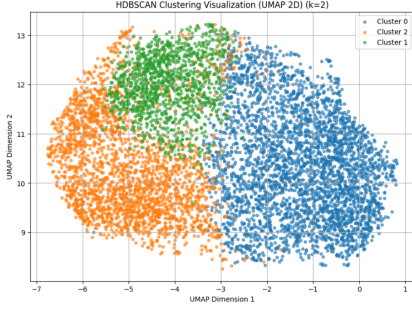


Figure 14

When allowing three clusters, HDBSCAN uncovers a new intermediate group, highlighting the model’s capacity to detect subtle variations in lyrical themes under relaxed density constraints.

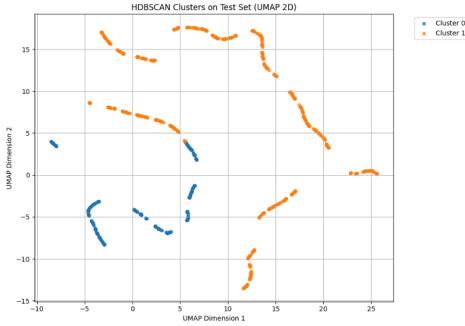


Figure 15

However, on the test set, HDBSCAN reverts to two clusters, underscoring its adaptive nature: the number of clusters is derived from the data itself, not predefined k .

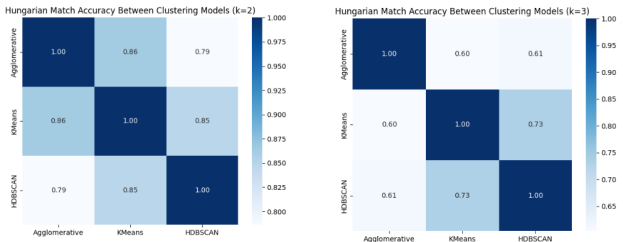


Figure 16

Using the Hungarian algorithm, agreement between models is strongest under the 2-cluster configuration, with 86.3% alignment between Agglomerative and K-Means, and 84.9% between K-Means and HDBSCAN. In contrast, the 3-cluster setting sees reduced alignment (60–73%), suggesting more divergence in how models interpret finer thematic divisions. K-Means and HDBSCAN maintain the highest consistency across both configurations.

VI. DISCUSSION

This study compared hierarchical (Agglomerative), partitioning (K-Means), and density-based (HDBSCAN) clustering on Sentence-BERT embeddings of English song lyrics. Across methods, a binary clustering ($k=2$) consistently yielded the most coherent and well-separated

results, based on Silhouette, Calinski–Harabasz, and Davies–Bouldin scores. This confirms the robustness of a two-cluster structure capturing the main thematic divide in the dataset.

Qualitative analysis using t-SNE (Figures 3–14) and UMAP shows that Cluster 0 typically groups narrative-driven lyrics with storytelling elements, while Cluster 1 contains motif-heavy, emotionally repetitive content. A ternary clustering ($k=3$) further reveals a transitional subtheme blending both styles, useful for finer-grained applications like playlist generation or discography analysis.

Among the methods, HDBSCAN achieved the highest binary Silhouette score (0.627), indicating strong performance in identifying dense, meaningful clusters. However, its sensitivity to hyperparameters and tendency to assign noise labels poses integration challenges. In contrast, K-Means and Agglomerative models offered more stable, noise-free clusters, making them suitable for scalable, real-time applications.

Challenges included the variability of lyrics, uneven cluster densities, and the impact of distance metrics and linkage methods (e.g., Ward–Euclidean vs. complete–cosine) on hierarchical clustering. SBERT embeddings, while powerful, also struggled with figurative language and cultural nuance—suggesting future enhancements like domain fine-tuning or multimodal fusion.

Overall, SBERT embeddings paired with clustering techniques can uncover meaningful lyrical patterns. A two-cluster solution clearly separates narrative and motif-driven lyrics, while a three-cluster option introduces intermediate themes. HDBSCAN offers high separation at the cost of complexity; K-Means and Agglomerative strike a balance between clarity and operational ease. Future directions include hybrid clustering, user validation, and integration into recommendation pipelines.

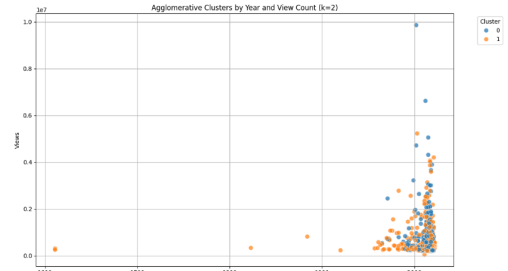


Figure 17

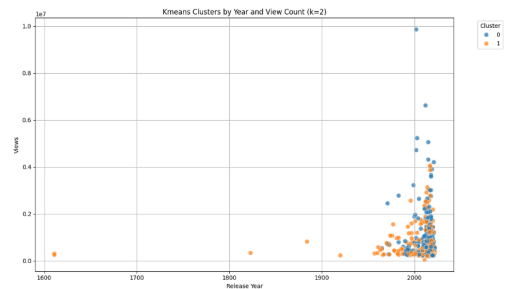


Figure 18

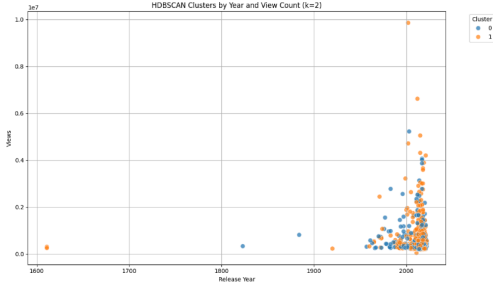


Figure 19

Visual analysis of the test set reveals that HDBSCAN clusters are distributed uniformly across release years and view counts, highlighting its density-driven nature. K-Means and Agglomerative, however, show stronger alignment with popularity trends—Cluster 0 in both models contains more high-view, post-2000 songs.

In terms of average view counts ($k=2$), K-Means showed the largest gap (766,163 vs. 571,062), followed by Agglomerative (739,936 vs. 604,800), effectively separating popular from less popular songs. HDBSCAN, interestingly, reversed this pattern, with Cluster 1 showing higher views, underscoring its data-driven boundary formation.

Median release years across clusters converged around 2015–2016 in the binary setup, indicating temporal features are less influential in two-cluster models. At $k=3$, Agglomerative displayed clearer temporal stratification (2012, 2015, 2016), with K-Means showing moderate separation. HDBSCAN maintained tight clustering around 2015–2016, reflecting its emphasis on local density.

Vocabulary analysis reveals a clear semantic separation within the data. Utilizing TF-IDF scores, there are distinct lexical patterns that define a consistent, binary thematic split across all clustering models. One cluster is characterized by its explicit and assertive nature, dominated by street slang, profanity, and confrontational terms (e.g., “nigga,” “bitch,” “fuck”). This group represents a raw, unfiltered narrative style often found in modern hip-hop. In stark contrast, the second cluster contains emotional and introspective content, defined by words related to love, vulnerability, and existential reflection (e.g., “love,” “heart,” “life”). This group captures more abstract and universally relatable themes. This fundamental split was independently discovered by Agglomerative, K-Means, and HDBSCAN, validating the models’ ability to uncover meaningful, interpretable groupings based purely on lyrical content.

VII. DEPLOYMENT

The proposed clustering system has strong potential for real-world deployment, particularly in digital music platforms such as Spotify, Joox, or SoundCloud. By precomputing SBERT embeddings, the system can cluster lyrics efficiently and periodically as new content is added. These clusters can enhance features like automated playlist generation, lyric-based recommendations, and semantic song discovery. However, ethical considerations remain important, including potential bias in language models and privacy concerns for user-submitted data. This system can enrich lyric-focused user experiences while remaining practical for deployment at scale.

VIII. CONCLUSION

This study clustered 10,000 song lyrics from genius.com using SBERT embeddings with Agglomerative, K-Means, and HDBSCAN algorithms. Both Agglomerative and K-Means performed best with a binary partition ($k=2$), yielding modest silhouette scores of ≈ 0.042 and ≈ 0.061 , respectively. In contrast, HDBSCAN identified a more robust two-cluster solution with a significantly higher silhouette score (≈ 0.627) and fewer noise points, highlighting the effectiveness of density-based methods for this data.

Increasing the cluster count beyond two consistently resulted in poorer internal validation scores across all algorithms. However, a three-cluster solution ($k=3$) did reveal finer subthemes, such as narrative-driven and motif-driven lyrical patterns. These results successfully address the study’s primary research questions regarding algorithm performance, hyperparameter impact, and the identification of latent thematic groupings.

The SBERT-based pipeline successfully uncovered coherent thematic groupings. External validation confirmed these clusters align with human perception, demonstrating practical value for recommendation systems and musicology research. Future work should focus on refining cluster interpretability through interactive visualizations and extending the methodology to multilingual corpora for more nuanced musical discovery.

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